

Echos of the Crown: Is there an Effect of Royal Speeches on Parliamentary Discussion?

Albert Lamb, Andre Garcia, Firas Abadi

July 29, 2026

Abstract

This paper studies whether the monarch’s annual address to parliament shapes legislative debate, using two measures of influence applied to the full text of UK parliamentary proceedings. The first model tracks whether vocabulary distinctive to the royal speech appears at elevated rates in parliamentary debate after the address. The second measures whether daily parliamentary attention shifts toward the topics the speech emphasizes, using a Continuous Interrupted Time Series design that separately identifies anticipation, immediate response, and anchoring effects. We find no evidence of vocabulary uptake. On topic attention, parliamentary debate shifts toward speech topics in the two weeks before the ceremony, there is no discontinuous jump on the day itself, and attention stabilizes at a higher level in the weeks that follow. The royal speech shapes what parliament talks about but not how it talks.

1 Introduction

Does the monarch’s addresses to parliament actually change what legislators talk about? Royal speeches are delivered at key moments in the parliamentary calendar, setting out priorities, marking occasions, and responding to events. Across these addresses, the monarch’s voice enters the public record repeatedly and visibly. If parliamentary debate shifts toward the topics emphasized in these speeches, that is evidence of real agenda-setting influence. If it does not, the ceremonial role of the monarchy is largely rhetorical.

The difficulty is that the speech does not arrive unexpectedly. Journalists speculate about its contents for weeks beforehand, ministers discuss priorities through public statements, and experienced parliamentarians know which issues are likely to feature. By the time the monarch speaks, parliament may have already begun adjusting its attention. A simple before-and-after comparison would confuse this anticipation with the effect of the speech itself.

This paper studies the question using two approaches. The first asks whether vocabulary distinctive to the royal speech subsequently appears at elevated rates in parliamentary debate. The second asks whether parliamentary attention shifts toward the speech topics, and whether this shift follows a predictable temporal pattern around the event. Together the two models provide a two-dimensional picture of royal influence: one measuring whether parliament starts speaking like the monarch, and one measuring whether parliament starts talking about the same things. A parliament that discusses the monarch’s topics without adopting the monarch’s words would suggest the speech shapes the agenda but not the rhetoric. We find exactly this pattern.

2 Theoretical Framework & Literature Review

When parliament debates one issue, it cannot simultaneously debate another. Floor time is finite, and placing one topic on the agenda necessarily displaces others [JB05, Kin84]. This zero-sum structure means that shifts in legislative attention are not just informative about what parliament prioritizes, but also about what it does not. Agenda-setting research has used this insight to model how attention moves across issues over time, typically finding that legislative priorities are slow to change and respond mainly to large, unexpected disruptions [BJ93].

The annual royal address fits a different pattern. Unlike a crisis or an unexpected political shock, the speech is a fixed point in the parliamentary calendar. Its date is known months in advance, and its broad themes are frequently anticipated through press coverage, government signaling, and the usual patterns of the legislative session. When an event is highly predictable, rational political actors do not wait for it to happen before responding. They begin adjusting in advance. [MR15] show formally that this anticipatory behavior must be treated as a substantive finding rather than a methodological nuisance, as ignoring it produces biased estimates of the event’s true effect.

Prior research confirms that Royal speeches influence what legislatures subsequently do [BJJ11, HK05], and work focused specifically on the Queen’s Speech shows that parliamentary attention shifts in response to its content [JJ09, JJ10]. The limitation of this literature is that it measures effects at the level of full parliamentary sessions

or annual periods. At that resolution, it is impossible to tell whether parliament was already moving toward the speech topics before the ceremony, whether the speech itself caused a sudden jump, or whether the effect only materialized gradually in the weeks after. Distinguishing these patterns requires daily data.

We therefore construct a daily panel of parliamentary attention and apply two measurement strategies. The first tracks whether distinctive vocabulary from the royal speech subsequently appears in parliamentary debate. The second uses topic modeling [BNJ03] to measure daily attention shares across policy areas, following the validation in [GS13] that automated text analysis methods can recover substantively meaningful and reliable signals from political speech when applied with appropriate care. To estimate whether the speech changes the trajectory of parliamentary attention, we use a Continuous Interrupted Time Series design [BT75]. Standard before-after comparisons are inadequate here because parliamentary debate on any given topic is relatively steady from one day to the next [BDM04, Lin15], and the CITS framework accounts for this by modeling the underlying trend in attention, allowing us to identify what happens before the speech, on the day itself, and in the weeks that follow.

3 Data & Measurement

3.1 Parliamentary Data

To construct the daily parliamentary panel, we utilized the digitized Hansard transcripts of the UK House of Commons, accessed via the [myS26] parliamentary data repository. We extract the complete daily debate record spanning from January 1, 1952, to December 31, 2025, providing a comprehensive longitudinal corpus of legislative behavior.

The data acquisition pipeline operates in two automated stages: retrieval and parsing. First, we programmatically query the repository to download the raw XML transcript for every active legislative day. Second, we parse the hierarchical XML structure to extract both the substantive text and the structural metadata of the debate.

Because parliamentary transcripts are organized topically, the parsing algorithm tracks changes in major and minor XML headings to assign a structural `section` label to each speech. We iterate through individual `<speech>` tags, flattening the nested paragraph elements into a single continuous text string for each intervention. Crucially, important metadata was extracted during this process: speaker name, unique member IDs, timestamps, and column numbers),

The final parsed outputs are appended sequentially into a continuous longitudinal dataset, transforming the raw daily XML files into the structured, document-level panel required for the LDA topic modeling and subsequent time-series analysis.

3.2 Royal Data

To construct the Royal panel, we utilized the selenium package to scrape the British Royal Family Speeches Archive. We used the, native to the website, filter to only scrape speeches from monarchs starting from 1952 (From the start of Elizabeth II reign - Present King Charles III reign). For each speech we also had the speaker name, date, title, and website link, although we did not use the link or title data in our analysis. The final raw output of royal speeches was then ordered chronologically.

3.3 Text Preprocessing and Topic Modeling

For model 1 each document is passed through spaCy’s `en_core_web_sm` model, which applies lemmatisation, followed by stopword removal and punctuation stripping. Tokens shorter than three characters are also removed, as these are typically function words and abbreviations

We choose lemmatisation over stemming for two reasons. First, lemmatisation produces morphologically coherent output (*running* → *run*), which matters for vocabulary matching in Model 1. Second, the TF-IDF scores used to extract distinctive royal vocabulary are more interpretable when tokens correspond to actual dictionary words rather than truncated stems.

For model 2 each document is passed through spaCy’s `en_core_web_sm` model, which applies stemming, followed by stopword removal and punctuation stripping. Tokens shorter than three characters are also removed, as these are typically function words and abbreviations

We then use Latent Dirichlet Allocation (LDA) [BNJ03] to measure daily parliamentary attention. This approach resolves three specific measurement problems in longitudinal text analysis.

Inductive Topic Discovery: Static dictionaries degrade over multi-decade timelines and fail to capture shifting vocabularies. LDA is an unsupervised model that identifies latent topics based on statistical word co-occurrence, grounding the topic categories in the actual vocabulary used during debates rather than researcher-defined codebooks.

Mixed-Distribution Classification: Parliamentary speeches frequently cover multiple issues. Algorithms that assign mutually exclusive categories introduce measurement error by overestimating primary topics and dropping secondary ones. LDA estimates a probability distribution over topics for each document, capturing proportional attention across multiple policy areas.

Dimensionality Reduction: Event-study frameworks require a continuous time-series variable. Raw text data forms a sparse, high-dimensional matrix. LDA reduces this dimensionality by projecting the vocabulary into a smaller set of latent topic variables. Aggregating these document-level probabilities to the daily level produces the continuous, dense measure of attention required for the regression models.

The choice of background filter size ($K = 200$ most frequent parliamentary terms) and the number of TF-IDF terms extracted per royal quarter ($N = 100$) were selected after sensitivity analysis (not too large that we consider the whole royal speech, but large enough so there was actually some words being filtered out). Reducing K to 100 leaves more terms in the surviving vocabulary but increases confounding from terms already commonplace in parliamentary debate.

4 Methodology

The two models use different text analysis methods by design. Model 1 uses TF-IDF to identify distinctive royal vocabulary because TF-IDF is suited to extracting terms that characterize a specific document relative to a corpus, which is what we need to isolate the agenda of a particular speech from background royal language. Model 2 uses LDA because the research question requires a continuous measure of parliamentary attention across policy topics. LDA produces a probability distribution over topics for each document, which can be aggregated to the daily level and tracked over time. So, TF-IDF measures lexical distinctiveness at the word level, while LDA measures thematic structure at the document level. Using both allows us to test whether royal influence operates at the level of words, topics, or neither.

4.1 Model 1: Vocabulary Uptake

4.1.1 Model Selection

To estimate whether royal speeches causally increase the penetration of their distinctive vocabulary into parliamentary debate, we use a panel regression with royal-quarter fixed effects, estimated on the uptake-score panel constructed in the pipeline below. We exploit the temporal symmetry of the observation window: for each royal speech quarter we observe parliamentary uptake in the two quarters before and the two quarters after the address, using the pre-speech observations as the within-unit counterfactual.

This identification strategy is similar in spirit to the internal control group logic employed in Model 2, where the zero-sum nature of parliamentary floor time allows topics omitted from the speech to serve as the control. Here, the pre-speech uptake observations serve an equivalent function: they estimate the baseline rate at which the royal surviving vocabulary appears in parliamentary debate absent any post-speech influence, allowing us to identify any incremental effect of the address itself.

4.2 Vocabulary Uptake Pipeline (Model 1)

Model 1 measures whether vocabulary that is distinctive to royal speeches (with respect to TF-IDF scoring) subsequently appears at elevated rates in parliamentary debate. The pipeline translates raw speech transcripts into a quarterly panel of *uptake scores*, each of which quantifies the share of parliamentary word tokens in a given quarter that can be attributed to the filtered vocabulary of a nearby royal speech quarter.

Step 1 — Corpus partitioning and temporal filtering. The raw transcripts are partitioned into a royal corpus and a parliamentary corpus. Both corpora are filtered to retain only speeches delivered from 2002 onward due to sparse data.

Step 2 — Quarterly aggregation. Individual speeches within each corpus are grouped by calendar quarter. All preprocessed tokens from speeches falling within the same three-month period are concatenated into a single quarterly document. We did this so that we did not have overlapping windows.

Step 3 — Text preprocessing. Each quarterly document is passed through a uniform preprocessing pipeline, applying: i) punctuation removal via regular expression, (ii) stopword removal using a standard English stopwords list, and (iii) lemmatization

Step 4 — TF-IDF extraction of royal vocabulary. A Term Frequency–Inverse Document Frequency (TF-IDF) model is fitted across all royal quarterly documents, with the vocabulary restricted to terms appearing in at least 1 % and no more than 85 % of quarters and capped at 10,000 features. For each royal quarter q , the K highest-scoring terms by TF-IDF weight form the candidate vocabulary set $\tilde{V}(s_q)$. This set contains the words that are most distinctive for royal quarter.

Step 5 — Parliamentary background estimation. For each royal quarter q , we identify the N most frequent terms in the parliamentary document from the quarter two periods prior, p_{q-2} , using a count vectoriser. This prior-window parliamentary vocabulary, denoted B_q , serves as an estimate of the baseline parliamentary vocabulary.

Step 6 — Background subtraction and vocabulary filtering. The candidate royal vocabulary is filtered by removing any term that also appears in the background set:

$$V(s_q) = \tilde{V}(s_q) \setminus B_q.$$

The intuition is straightforward: if a word already featured prominently in parliamentary debate two quarters before the royal address, its subsequent appearance in parliament cannot plausibly be attributed to the speech. Removing such terms from $\tilde{V}(s_q)$ reduces confounding from shared topical trends that arise independently in both institutions.

Step 7 — Uptake score computation. For each royal quarter q and each temporal offset $k \in \{-2, -1, +1, +2\}$, the uptake score is defined as the proportion of word tokens in the parliamentary quarterly document p_{q+k} that belong to the surviving vocabulary $V(s_q)$:

$$U(q, k) = \frac{|\{w \in p_{q+k} : w \in V(s_q)\}|}{|p_{q+k}|},$$

where $|p_{q+k}|$ denotes the total token count of the parliamentary document in quarter $q + k$ and the numerator counts, with multiplicity, all tokens in that document that appear in $V(s_q)$. Offsets $k \in \{-2, -1\}$ correspond to parliamentary quarters *preceding* the royal speech and constitute the pre-speech baseline; offsets $k \in \{+1, +2\}$ correspond to quarters *following* the speech and constitute the post-speech window of interest. The contemporaneous quarter ($k = 0$) is excluded because it is ambiguous whether parliamentary activity within that quarter precedes or follows the speech event.

Step 8 — Panel construction. The uptake scores computed in Step 7 are assembled into a balanced panel dataset. Each row corresponds to a unique (royal quarter, parliamentary quarter) pair and records the uptake score $U(q, k)$, the offset k , and a binary post-speech indicator $Post_k = \mathbf{1}[k > 0]$.

4.2.1 Model Specification

We estimate the following ordinary least squares specification on the uptake panel:

$$Uptake_{q,k} = \alpha + \beta_{post} \cdot Post_k + \beta_{offset} \cdot Offset_k + \sum_q \gamma_q \cdot \mathbf{1}[RoyalQuarter = q] + \varepsilon_{q,k},$$

where:

- $Uptake_{q,k}$ is the uptake score for parliamentary quarter $q + k$ relative to the surviving royal vocabulary of quarter q .
- $Post_k$ is a binary indicator equal to 1 for post-speech quarters ($k > 0$) and 0 for pre-speech quarters ($k < 0$).
- $Offset_k$ is the integer quarter offset k , entered as a continuous linear control to absorb any monotonic trend in uptake with temporal distance from the speech.
- $\mathbf{1}[RoyalQuarter = q]$ are dummy variables for each royal speech quarter, entered as a categorical fixed effect. These absorb all time-invariant characteristics of individual speech occasions, including political context, the monarch’s identity, and seasonal legislative patterns.
- $\varepsilon_{q,k}$ is the error term.

4.3 Model 2: Topic Attention

4.3.1 Daily Topic Aggregation

To construct the daily parliamentary panel, document-level topic proportions are weighted by word count to prevent brief procedural remarks from skewing the daily agenda. Let D_t be the set of all parliamentary speeches delivered on day t . Let N_d denote the total word count of document d , and let $\theta_{d,k}$ represent the proportion of document d allocated to topic k by the LDA model. The aggregate daily attention allocated to topic k on day t , denoted as $Y_{t,k}$, is calculated as:

$$Y_{t,k} = \frac{\sum_{d \in D_t} N_d \cdot \theta_{d,k}}{\sum_{d \in D_t} N_d} \quad (1)$$

4.3.2 The Dynamic 80% Rule and Treatment Assignment

The treatment group is recalculated dynamically for each year’s Royal speech (E). Let $\theta_{E,k}$ represent the proportion of the monarch’s speech E dedicated to topic k . We rank all K topics in descending order based on their weight in that specific speech:

$$\theta_{E,(1)} \geq \theta_{E,(2)} \geq \dots \geq \theta_{E,(K)} \quad (2)$$

The treatment group for event E , denoted as $\mathcal{T}_E$, consists of the top n topics required to capture 80% of the speech’s total attention. We define n as the smallest integer satisfying the cumulative threshold:

$$\sum_{i=1}^n \theta_{E,(i)} \geq 0.80 \quad (3)$$

Once the treated topics are isolated ($\mathcal{T}_E = \{(1), \dots, (n)\}$), we define the binary treatment indicator for topic k during the observation window surrounding event E :

$$Treated_{k,E} = \begin{cases} 1 & \text{if } k \in \mathcal{T}_E \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

4.3.3 Model Selection

To estimate the causal effect of the monarch’s speech on parliamentary discourse, we use a Comparative Interrupted Time Series (CITS) design. A more traditional approach like Difference-in-Differences (DiD) relies on parallel pre-trends and compares static averages between pre-treatment and post-treatment periods. However, as pointed out by [MR15], rational actors frequently anticipate policy changes. If the royal institution actively signals its agenda in the week leading up to the ceremony, a standard DiD will absorb this anticipatory behavior into the pre-treatment average, biasing the estimates in a predictable direction. The CITS framework corrects for this by treating time continuously. Instead of assuming static baseline averages, CITS models the pre-existing trajectory of both the treated and control groups, allowing us to measure the anticipatory momentum explicitly and test whether the speech altered it.

4.3.4 Model Specification

$$\begin{aligned} Attention_{c,t} = & \alpha + \beta_1 Treated_c + \beta_2 Post_t + \beta_3 Time_t + \beta_4 (Treated_c \times Post_t) + \beta_5 (Treated_c \times Time_t) \\ & + \beta_6 (Post_t \times Time_t) + \beta_7 (Treated_c \times Post_t \times Time_t) + \gamma_c + \delta_m + \zeta_y + \epsilon_{c,t} \end{aligned}$$

- $Attention_{c,t}$ is the proportion of parliamentary discourse dedicated to topic c on day t .
- $Treated_c$ is a binary indicator equal to 1 if the topic was emphasized in the monarch’s speech, and 0 if it was omitted (the control group).
- $Post_t$ is a dummy variable equal to 1 for days following the speech, and 0 for days preceding it.
- $Time_t$ is a continuous running variable representing the number of days relative to the speech (Day 0).
- $\gamma_c, \delta_m, \zeta_y$ represent topic, month, and year fixed effects, respectively, to control for baseline topic popularity and seasonal legislative cycles.
- $\epsilon_{c,t}$ are standard errors, clustered to account for serial correlation within topics over time.

The mechanisms of interest are captured by:

- β_5 : the **Anticipation Effect** — the difference in pre-trend slopes between treated and control topics.
- β_4 : the **Day 0 Shock** — the immediate vertical shift in attention on the day of the speech.
- β_7 : the **Anchoring Effect** — the change in the trajectory of attention following the ceremony.

4.3.5 Identification Strategy and Unobserved Confounders

Zero-sum agenda space and the internal control group. Observational models of legislative behavior are vulnerable to unobserved confounding variables. Our identification strategy addresses this by exploiting the natural constraints of the legislative space. Parliamentary floor time is a finite, zero-sum resource [JB05, Kin84], meaning that attention given to one policy area necessarily crowds out space for another.

By using the policy topics omitted from the monarch’s speech as our internal control group, we measure a causal premium relative to the baseline trajectory of omitted topics, effectively absorbing background noise and addressing three primary threats to identification:

1. **Exogenous Macro-Shocks.** A sudden crisis occurring within our 28-day window would disrupt the legislative calendar. Because legislative space is zero-sum, the shock would reduce floor time for both treated and control topics equally. Modelling the difference between the two groups removes this shared effect.
2. **Slow-Moving Shifts.** Confounders such as macroeconomic cycles, partisan composition, or long-term institutional changes could influence baseline parliamentary activity. By restricting the observation window to the 28 days around the ceremony, these slow-moving variables remain essentially constant.
3. **Endogenous Media Pressure.** A classic endogeneity concern is whether the monarch is setting the agenda or simply responding to topics already dominating the news cycle. Our CITS model addresses this by explicitly modelling the pre-speech trend. By estimating this baseline and interacting it with the post-speech window, we isolate the effect that is specific to the speech itself.

5 Results & Discussion

5.1 Model 1

The table and graph below presents the panel regression estimates for vocabulary uptake. The results indicate that, conditional on royal-quarter fixed effects and the linear offset control, there is no statistically significant increase in the share of surviving royal vocabulary in parliamentary debate following a royal speech.

Table 1: OLS Results — Real V(s)

Variable	Coef.	Std. Err.	95% CI	p-value
post	+0.000339	(0.000608)	[−0.000853, 0.001531]	0.578
offset	+0.000071	(0.000226)	[−0.000372, 0.000515]	0.753
$N = 98$				
$R^2 = 0.9311$				
Adj. $R^2 = 0.9059$				
$F\text{-stat} = 249.2142, p < 0.001$				

Notes: Dependent variable: uptake. Robust standard errors (HC1) in parentheses. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $^\dagger p < 0.1$.

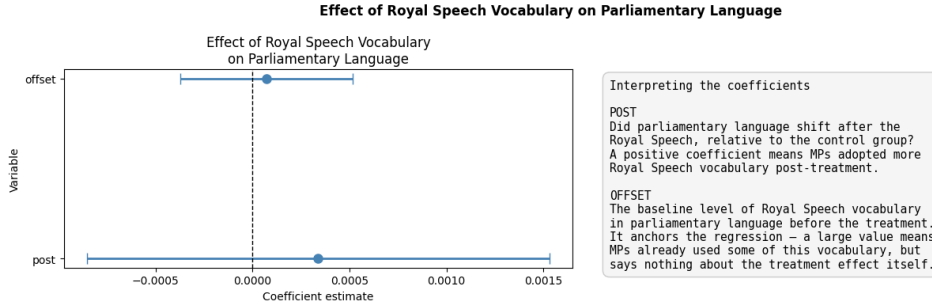


Figure 1: Effect of Royal Speech Vocabulary on Parliamentary Language. Coefficient estimates with 95% confidence intervals. The dashed vertical line denotes zero.

5.1.1 Interpretation

The absence of a detectable uptake effect admits two non-exclusive interpretations. First, royal speeches may genuinely exert no influence on parliamentary vocabulary at the quarterly level. The surviving vocabulary sets $V(s_q)$, though filtered to remove shared background terms, may still contain words that both corpora adopt independently in response to the same political environment, such that the background subtraction does not fully eliminate confounding. Alternatively, any genuine royal influence on parliamentary language may operate through channels not captured by raw term-frequency measures, for instance, through the framing, sequencing, or weighting of topics rather than through the introduction of novel vocabulary.

5.2 Model 2

Table 2 presents the CITS estimates for parliamentary attention. The results indicate that the monarch’s speech does not act as an isolated shock to the legislative agenda. Instead, attention shifts through a multi-week process, which the model captures in three distinct phases: anticipation, culmination, and anchoring.

5.2.1 The Anticipation Effect

The model identifies a positive and highly significant anticipation effect ($\text{treated:time} = 0.0006, p < 0.01$). In the 14 days preceding the speech, parliamentary attention systematically shifts toward the topics that will be included in the address. Because plenary time is a zero-sum resource, this upward pre-trend demonstrates that speech topics crowd out competing issues well before the event occurs. This shift likely reflects anticipation driven by routine political forecasting, media coverage, and the standard legislative calendar prior to a new session.

5.2.2 The Day 0 Culmination

The coefficient for the immediate Day 0 shock (treated:post) is statistically insignificant ($0.0011, p = 0.547$). The data shows no sudden jump in attention on the day the speech is delivered. In the context of the CITS framework, this suggests that the thematic contents of the speech are already reflected in the chamber’s discourse by the time the monarch speaks. The two-week anticipation phase has already shifted the agenda, and the ceremony acts as a formal culmination of this build-up rather than a source of new information.

5.2.3 The Anchoring Effect

Following the speech, the upward pre-trend halts. The negative, highly significant triple interaction term ($\text{treated:post:time} = -0.0008, p < 0.01$) captures this structural break. Because attention was elevated during the pre-speech window, this break does not represent a return to the baseline. Instead, attention stabilizes at a higher level. The event serves as an anchor, halting the movement of the pre-trend and locking the agenda in place for the following weeks.

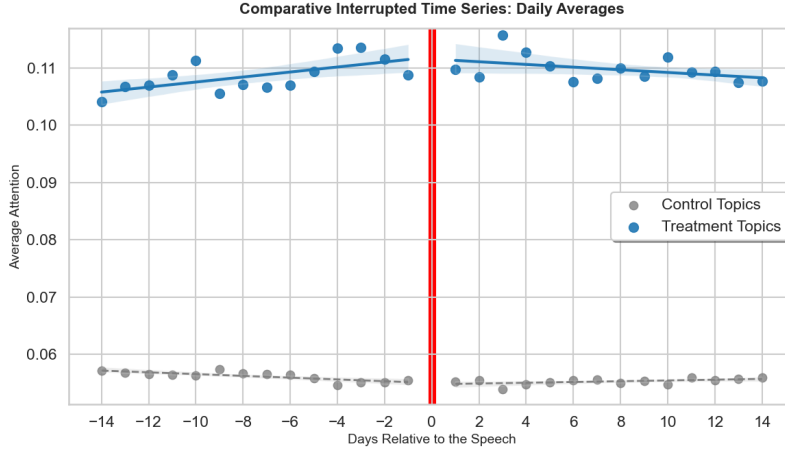


Figure 2: This figure plots the daily average parliamentary attention over a 14-day window before and after the executive speech (marked by the solid red vertical line at Day 0). The solid blue line and markers represent the treatment group (the core topics constituting the executive’s agenda), while the dashed grey line represents the control group (the remaining topics).

Table 2: Comparative Interrupted Time Series (CITS) Estimates on Parliamentary Attention

Variables	Coefficient (HAC Std. Error)
Main CITS Effects	
Anticipation Effect (<i>treated:time</i>)	0.0006*** (0.000)
Day 0 Shock (<i>treated:post</i>)	0.0011 (0.002)
Anchoring Effect (<i>treated:post:time</i>)	−0.0008*** (0.000)
Baseline Dynamics	
Treated Group Baseline (<i>treated</i>)	0.0041*** (0.001)
Post-Speech Baseline (<i>post</i>)	−0.0003 (0.001)
Time Trend Baseline (<i>time</i>)	−0.0001** (0.000)
Post-Speech Trend (<i>post:time</i>)	0.0002* (0.000)
Intercept	0.0370*** (0.008)
Observations	77,280
R^2	0.588
Adjusted R^2	0.587
Topic Fixed Effects	Yes
Month Fixed Effects	Yes
Year Fixed Effects	Yes

Note: HAC standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Robustness Checks

6.1 Model 1: Placebo Vocabulary Test

To test whether royal speech vocabulary uptake reflects genuine rhetorical influence rather than background parliamentary trends, we construct a placebo vocabulary $V_{\text{placebo}}(s)$ by randomly sampling $n = 100$ terms from the parliamentary debate corpus of the preceding quarter $q - 1$, excluding terms already in $V(s)$. These words were in active parliamentary use before the speech but are unrelated to its distinctive content.

We compute uptake of $V_{\text{placebo}}(s)$ over the same ± 2 quarter window and run an identical regression, with uptake defined as the share of parliamentary words belonging to $V_{\text{placebo}}(s)$.

Table 3: Event-study regression results: effect of royal speech vocabulary on parliamentary uptake. Real $V(s)$ uses TF-IDF distinctive terms filtered against background parliamentary vocabulary. Placebo $V(s)$ uses randomly sampled parliamentary terms. Standard errors in parentheses. Quarter fixed effects included in all specifications. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	Real $V(s)$	Placebo $V(s)$
post	-0.000082 (0.000255)	-0.000174* (0.000096)
offset	0.000074 (0.000098)	0.000057** (0.000028)
R^2	0.8400	0.9888
Adj. R^2	0.7814	0.9846
N	98	98
Quarter FE	Yes	Yes

The placebo test confirms the null result for real $V(s)$ is meaningful. The placebo specification detects a significant negative $\hat{\beta}_{post}$, indicating that parliamentary language systematically moves *away* from recently used vocabulary following a royal speech, which is consistent with natural opic-switching rather than vocabulary absorption. As the placebo picks up this background signal while the real $V(s)$ specification does not suggest the measure is sensitive enough to detect an effect if one existed; the absence of a significant coefficient on real $V(s)$ therefore reflects a genuine null result.

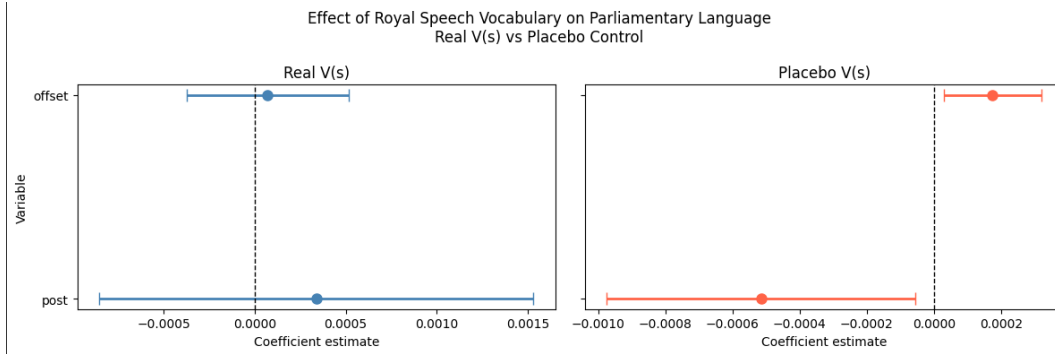


Figure 3: Effect of Royal Speech Vocabulary on Parliamentary Language: Real $V(s)$ vs Placebo Control. Coefficient estimates with 95% confidence intervals. The dashed vertical line denotes zero.

6.2 Model 2: Robustness of CITS Estimates

We conduct three tests to verify the CITS estimates against spurious timing, serial correlation, and window sensitivity.

6.2.1 Temporal Placebo Tests (Fake Dates)

To ensure the observed phase shifts are tied to the actual ceremony, we run a permutation test on the event dates. Over 140 iterations, we randomly shift Day 0 by 30 to 200 days for each speech, reconstruct the ± 14 -day panel, and re-estimate the model. The empirical p-values (Anticipation Effect $p = 0.0071$, Anchoring Effect $p = 0.00$) confirm that the true coefficients fall outside the placebo distribution. The observed trends do not occur on non-event days.

6.2.2 Topic Placebo Tests (Permutation in Space)

To account for over-rejection rates caused by panel serial correlation [BDM04, Lin15], we permute the treatment assignment. Over 140 iterations, we randomly assign treated status to topics in the control group, maintaining the true ratio of treated to untreated topics, and re-run the regression. The true CITS coefficients fall strictly outside the resulting placebo distribution ($p = 0.00$), confirming that the agenda-setting effect is specific to the topics explicitly prioritised in the speech.

6.2.3 Window Sensitivity

Our primary specification uses a ± 14 -day window to isolate the event while holding slow-moving institutional variables constant. We re-estimate the model using alternative bandwidths of ± 7 , ± 10 , ± 21 , and ± 28 days. The core coefficients remain stable in both direction and statistical significance across all windows.

Table 4: Window Sensitivity Results for Anticipation and Anchoring Effects

Window	Anticipation	Anticipation <i>p</i> -val	Anchoring	Anchoring <i>p</i> -val
± 7 days	0.00142	0.000	-0.00156	0.004
± 10 days	0.00093	0.000	-0.00094	0.012
± 14 days	0.00061	0.000	-0.00083	0.000
± 21 days	0.00061	0.000	-0.00083	0.000
± 28 days	0.00061	0.000	-0.00083	0.000

7 Limitations

7.1 Model 1

Model 1 faces several limitations regarding temporal resolution and measurement precision. First, aggregating speeches quarterly prevents overlapping observation windows but masks immediate, within-quarter variation. This temporal bluntness likely yields a conservative estimate of short-term executive influence. Second, the uptake score captures the raw frequency of surviving royal vocabulary but cannot distinguish between supportive adoption and critical opposition. It measures baseline usage, not policy endorsement. Finally, the two-quarter background subtraction may fail to completely filter out vocabulary driven by slow-moving macroeconomic or political trends rather than the speech itself, potentially inflating the uptake metric.

7.2 Model 2

Model 2 presents its own limitations regarding measurement, generalization, and causal mechanisms. Like Model 1, the Continuous Interrupted Time Series (CITS) dependent variable measures the raw volume of parliamentary attention allocated to specific topics, rather than policy agreement or the tone of the debate. Furthermore, while the model precisely isolates temporal shifts in the legislative agenda, it cannot identify the specific behavioral drivers—such as media forecasting, political speculation, or party discipline—causing those shifts. Finally, these findings rely on the highly centralized, zero-sum scheduling of a Westminster parliament. The observed anticipation and anchoring effects may not necessarily generalize to systems with separated powers or decentralized committee structures.

8 Conclusion

This paper was designed to answer a simple question: Is there an Effect of Royal Speeches on Parliamentary Discussion? The answer depends on how influence is measured.

The two models approach the question from different angles. The first asks whether the monarch’s words spread into parliamentary language, treating rhetoric as a direct channel of influence. The second asks whether the monarch’s themes spread into parliamentary attention, treating agenda allocation as the channel. A null result in the first and a positive result in the second suggests that royal influence operates at the level of what parliament talks about, not how it talks.

On vocabulary, we find no evidence that parliament adopts the distinctive language of the royal speech following the address. The surviving vocabulary sets, carefully filtered to remove terms already present in parliamentary debate, show no statistically significant increase in uptake after the speech. This null result is supported by a placebo test confirming that the measure is sensitive enough to detect background trends in parliamentary language, and that the absence of an effect in the main specification reflects a genuine absence of influence rather than a failure of measurement.

In terms of topic attention, we find that the parliamentary debate shifts to the subjects emphasized in the royal speech. In addition, the timing of this shift reveals something important about how the influence operates. The movement begins before the speech, not after it. In the two weeks preceding the address, attention gradually concentrates on the topics that will be emphasized, consistent with anticipation driven by media coverage and

political signaling. On the day of the speech itself, there is no sudden jump. By that point, the shift has already occurred, and the ceremony serves as a formal endpoint rather than a trigger. In the weeks that follow, the upward trend halts and attention settles at a higher level, suggesting the speech sets the agenda in place rather than continuously drive it.

Taken together, these findings suggest that the royal speech functions less as a shock to the legislative agenda and more as an anchor. Its influence operates through topic attention rather than language, and much of the observable effect precedes the event itself. This has broader implications for how we think about agenda-setting: the formal moment of a speech may matter less than the anticipatory process that surrounds it.

References

- [BDM04] Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275, 2004.
- [BJ93] Frank R. Baumgartner and Bryan D. Jones. *Agendas and Instability in American Politics*. University of Chicago Press, 1993.
- [BJJ11] Shaun Bevan, Peter John, and Will Jennings. Keeping party programmes on track: The transmission of the policy agendas of the speech from the throne to acts of the uk parliament. *European Political Science Review*, 3(3):395–417, 2011.
- [BNJ03] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3(Jan):993–1022, 2003.
- [BT75] George E. P. Box and George C. Tiao. Intervention analysis with applications to economic and environmental problems. *Journal of the American Statistical Association*, 70(349):70–79, 1975.
- [GS13] Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013.
- [HK05] Sara Binzer Hobolt and Robert Klemmensen. Responsive government? public opinion and government policy preferences in britain and denmark. *Political Studies*, 53(2):379–402, 2005.
- [JB05] Bryan D. Jones and Frank R. Baumgartner. *The Politics of Attention: How Government Prioritizes Problems*. University of Chicago Press, 2005.
- [JJ09] Will Jennings and Peter John. The dynamics of political attention: Public opinion and the queen’s speech in the united kingdom. *American Journal of Political Science*, 53(4):838–854, 2009.
- [JJ10] Will Jennings and Peter John. Punctuations and turning points in british politics: The policy agenda of the queen’s speech, 1940–2005. *British Journal of Political Science*, 40(3):561–586, 2010.
- [Kin84] John W. Kingdon. *Agendas, Alternatives, and Public Policies*. Little, Brown, 1984.
- [Lin15] Ariel Linden. Conducting interrupted time-series analysis for single- and multiple-group comparisons. *The Stata Journal*, 15(2):480–500, 2015.
- [MR15] Anup Malani and Julian Reif. Interpreting pre-trends as anticipation: Impact on estimated treatment effects from tort reform. *Journal of Public Economics*, 124:1–17, 2015.
- [myS26] mySociety. Theyworkforyou: Uk parliament data, 2026. Accessed and downloaded via XML data repository.